

Utility Systems and Power Failure

Policy Number	POL-EM-054	Owner / Department	Facilities / Environment of Care
Effective Date	2023-08-15	Version	2.4
Last Revised	2026-02-22	Next Review	2028-02-22
Approved By	Environment of Care Committee	Applies To	All staff in clinical and support areas; coordinated by Facilities and the House Supervisor

1. Purpose

To maintain safe patient care during failure of a critical utility system — electrical power, water, medical gas, HVAC, vertical transport, or communications — through an emergency power supply system and documented contingency procedures, consistent with The Joint Commission Environment of Care (utility systems) standards, NFPA 99, NFPA 110, and the CMS Emergency Preparedness Rule.

2. Scope

Applies to any interruption of a utility that supports patient care or a safe environment. Staff recognize the failure, protect patients dependent on the affected utility, and notify Facilities so the system can be restored or a backup engaged.

3. Definitions

- **Emergency Power Supply System (EPSS)** — the generator(s) and transfer switches that restore power to life-safety, critical, and equipment branches on loss of normal power.
- **Red (emergency) outlets** — receptacles powered by the emergency generator branch; life-support and critical equipment must be plugged into these.
- **Life-safety branch** — emergency lighting, exit signs, fire alarm, and medical-gas alarms.
- **Critical branch** — task lighting and receptacles serving patient care that is essential to life.

4. Procedure — Power Failure

IMPORTANT — On loss of normal power, the emergency generator restores power to red outlets within about 10 seconds. Immediately verify that all life-support and critical equipment (ventilators, monitors, pumps, suction) is on a red outlet, confirm it has restarted, and switch any device not on emergency power. Keep flashlights at each nurse station; never use elevators until cleared.

1. Ensure patient safety first: verify ventilators, monitors, and infusion pumps are running on emergency power; move any to red outlets if not.
2. For a device that did not restart, provide manual support (e.g., bag-valve ventilation) and call for help.
3. Notify Facilities / the plant operator and the House Supervisor; report the location and extent of the outage.

4. Deploy flashlights; secure elevators and verify no one is trapped; restrict non-essential elevator and equipment use.
5. If the outage is prolonged or widespread, the House Supervisor considers HICS activation and surge/transfer planning.
6. File an event report per POL-RM-013 and log the generator run and restoration time.

5. Severity & Reporting

Utility failures are classified under POL-RM-013. A failure that harms a patient, or a complete loss of emergency power or of a utility to a critical care area (ICU, OR, ED), is **SEV-1**: notify the Administrator-on-Call, Safety Officer, and Facilities leadership immediately and within 1 hour, and activate HICS for a prolonged or facility-wide loss. A localized failure requiring intervention without patient harm is **SEV-2** (within 4 hours). A brief interruption with automatic restoration and no impact, or a deficiency found on rounds, is **SEV-3 / Near Miss** within 24 hours.

6. Documentation Requirements

- Utility affected, location, and time of failure and restoration.
- Whether emergency power engaged automatically; generator run time.
- Patient-care equipment affected and any manual support provided.
- Injuries or clinical impact; notifications and HICS status.

7. References

- The Joint Commission, Environment of Care — utility systems (EC.02.05.01, EC.02.05.03, EC.02.05.07).
- NFPA 99, Health Care Facilities Code; NFPA 110, Emergency and Standby Power Systems.
- CMS Emergency Preparedness Rule, 42 CFR 482.15 (emergency and standby power).
- Related: POL-RM-013 (severity & reporting), POL-EM-055 (medical gas failure), POL-EM-053 (HICS / Code Triage).